Data Analytics for IoT Solutions

steps to analyze the IoT solution

Module VI

Data generation, Data gathering, Data Pre-processing, data analyzation, application of analytics, Exploratory Data Analysis

By

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Solve the Scenario

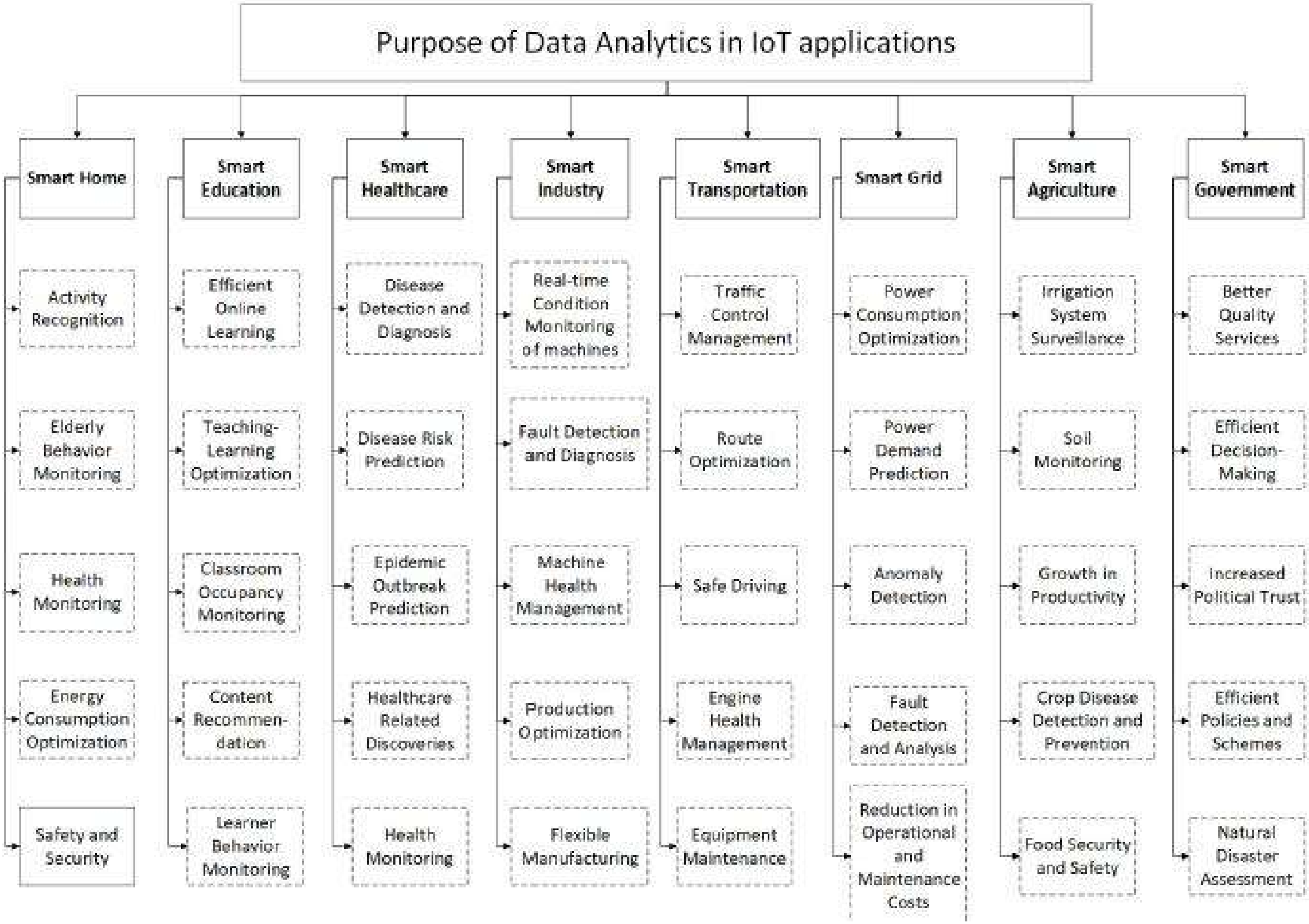
- Hunger reduction through smart farming, IoT analytics could be the monitoring and optimization of crop irrigation. The IoT sensors can be installed in the soil to measure moisture levels and transmit the data to a central hub. The hub uses analytics to compute the crop's water requirements based on crop type, soil type, and weather conditions. The system can then trigger smart irrigation systems to apply the right amount of water at the right time, reducing water waste and increasing crop yields. The data collected and analyzed can also help farmers to identify trends and improve crop management decisions.
- Discuss about data generation, Data gathering for the given example?
- Identify the data preprocessing techniques required for it, How IoT analytics can be defined for the above scenario.

Categories of IoT data

- **Status data**
  - whether an appliance is off or on, whether there are available spots at a property, etc
- **Location data**
  - location data for fleet management, asset tracking, employee monitoring
- **Automation data**
  - control devices inside a house, vehicles on the road, and other moving parts of any system
- **Actionable data**
  - an extension of status data
  - the system processes it and transforms into **easy-to-carry-out instructions**. Actionable data is often **used in forecasting** and **prediction, energy consumption** and workplace efficiency optimization, as well as during long-term decision-making

Benefits of IoT Data Collection

- connected devices and the Internet of Things making big data collection
- Healthcare :
  - healthcare professionals improve **the precision of diagnosis, as well as the speed of post-op recovery**.
  - ensure **safety inside the facility**, tracking both patients and staff.
  - improve **medication adherence**, monitor the effects of treatment, and prevent theft during shipping, as well as at the warehouse.
- Manufacturing:
  - create a **favorable environment for the peak performance of factory equipment**, monitor worker **productivity and integrate big data** performance monitoring solutions for predictive analytics and maintenance
- Agriculture:
  - monitor **farming sites forecast the likelihood** of natural disasters and their impact on crops sensor data helps **design efficient plant treatment**, monitor water usage, and reduce the amount of workforce needed to manage the site
- **Energy**
  - By **tracking the amount of electricity spent by the property**, facility managers become more aware of potential ways to reduce energy consumption
- **Smart homes :**
  - Security systems, **smart plugs, and other appliances all use IoT for data collection to ensure energy** efficiency, as well as in-house safety
- **Transportation :**
  - Traffic **congestion managers, virtual parking assistants**, fleet management tools, and fuel consumption monitoring devices are all sensor data collection examples



Purpose of Data Analytics in IoT applications
Smart Home
Activity Recognition
Elderly Behavior Monitoring
Health Monitoring
Energy Consumption Optimization
Safety and Security
Smart Education
Efficient Online Learning
Teaching-Learning Optimization
Classroom Occupancy Monitoring
Content Recommen-dation
Learner Behavior Monitoring
Smart Healthcare
Disease Detection and Diagnosis
Disease Risk Prediction
Epidemic Outbreak Prediction
Healthcare Related Discoveries
Health Monitoring
Smart Industry
Real-time Condition Monitoring of machines
Fault Detection and Diagnosis
Machine Health Management
Production Optimization
Flexible Manufacturing
Smart Transportation
Traffic Control Management
Route Optimization
Safe Driving
Engine Health Management
Equipment Maintenance
Smart Grid
Power Consumption Optimization
Power Demand Prediction
Anomaly Detection
Fault Detection and Analysis
Reduction in Operational and Maintenance Costs
Smart Agriculture
Irrigation System Surveillance
Soil Monitoring
Growth in Productivity
Crop Disease Detection and Prevention
Food Security and Safety
Smart Government
Better Quality Services
Efficient Decision-Making
Increased Political Trust
Efficient Policies and Schemes
Natural Disaster Assessment

1. Data Gathering

- **Data gathering in IoT (Internet of Things)** refers to the process of **collecting, processing, and transmitting data** from various IoT devices, sensors, and smart systems to extract meaningful insights. It is the foundation of IoT-based decision-making and automation.
- Will be performed at
  - **Sensors & Devices**
  - Edge Processing
  - **Communication Networks**
  - Cloud/Local Storage
  - Data Processing & Analytics
- It improves Decision-Making
- Predictive Maintenance
- **Smart Automation**
- **Optimized Resource Usage**

2. Data Creation or generation

- IoT data is the information collected by connected devices — sensors, wearables, and others.
- The IoT data is nothing but **data produced by those things**, the **new services** you can enable via those **connected things**, and convert them in to the **business insights** that the data can reveal.
  - Important data required for performing applications
- the world of IoT, the creation of massive amounts of data from sensors is common.
  - ***Modern jet engines*** are fitted with thousands of sensors that generate a whopping 10GB of data per second
- Need
  - demand fast data analytics with minimal delay.

Example : Data collection for Painting a house

Brand	EXT/INT	Type	Color(1/many)	Size	Days	Color change	No. of persons	Cost
								A
								d
								d

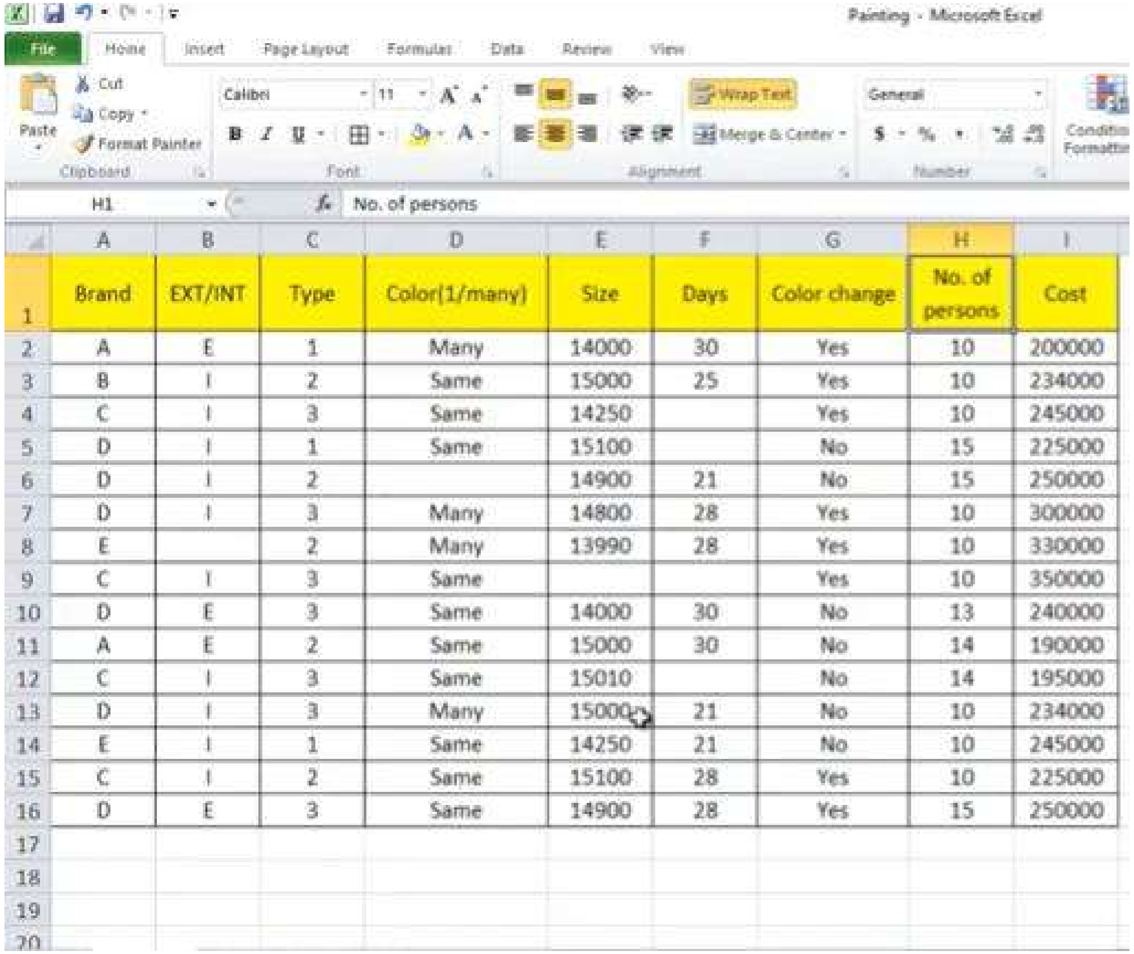
Operations on Data Set

- Creating a table
- Inserting values into the table
- Modify the values : drop, change groupby
  - Using relational databases.

Data Set

- Structured data by RDBMS
- Matrix form given by
  - Oracle DB, PhP, CSV,Excel
  - Column: domain of interest (characteristics, features
  - Row : experience of individual experience
    - Gains from each individual data entry by person or resource.
- What are you going do with data set
  - Data preprocessing.

Sample Dataset

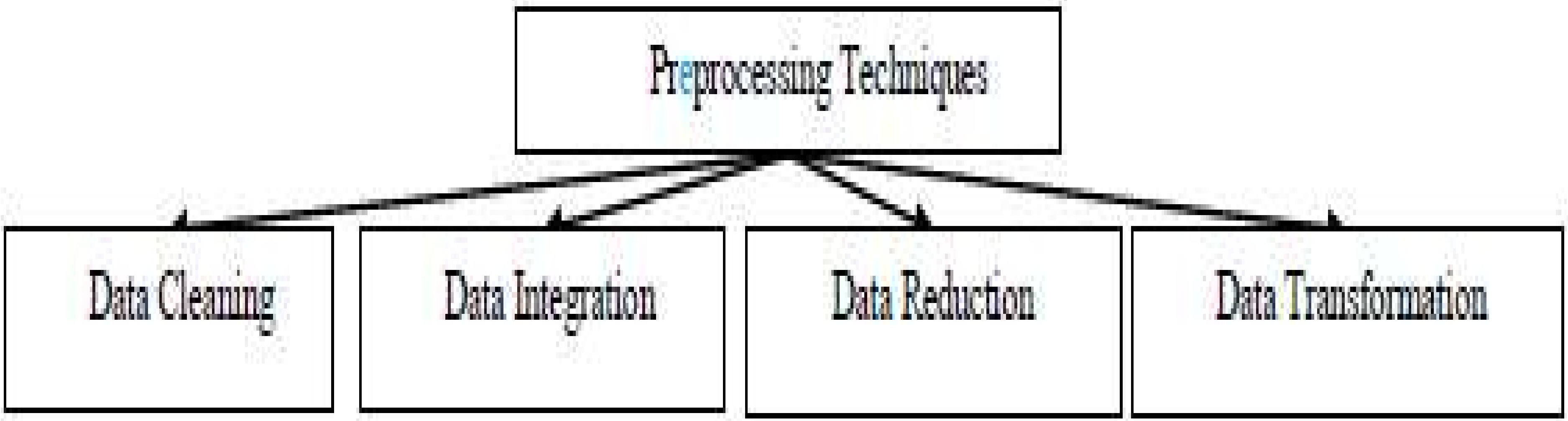


Brand	EXT/INT	Type	Color(1/many)	Size	Days	Color change	No. of persons	Cost
A	E	1	Many	14000	30	Yes	10	200000
B	I	2	Same	15000	25	Yes	10	234000
C	I	3	Same	14250		Yes	10	245000
D	I	1	Same	15100		No	15	225000
D	I	2		14900	21	No	15	250000
D	I	3	Many	14800	28	Yes	10	300000
E		2	Many	13990	28	Yes	10	330000
C	I	3	Same			Yes	10	350000
D	E	3	Same	14000	30	No	13	240000
A	E	2	Same	15000	30	No	14	190000
C	I	3	Same	15010		No	14	195000
D	I	3	Many	15000	21	No	10	234000
E	I	1	Same	14250	21	No	10	245000
C	I	2	Same	15100	28	Yes	10	225000
D	E	3	Same	14900	28	Yes	15	250000

3. Data pre-processing

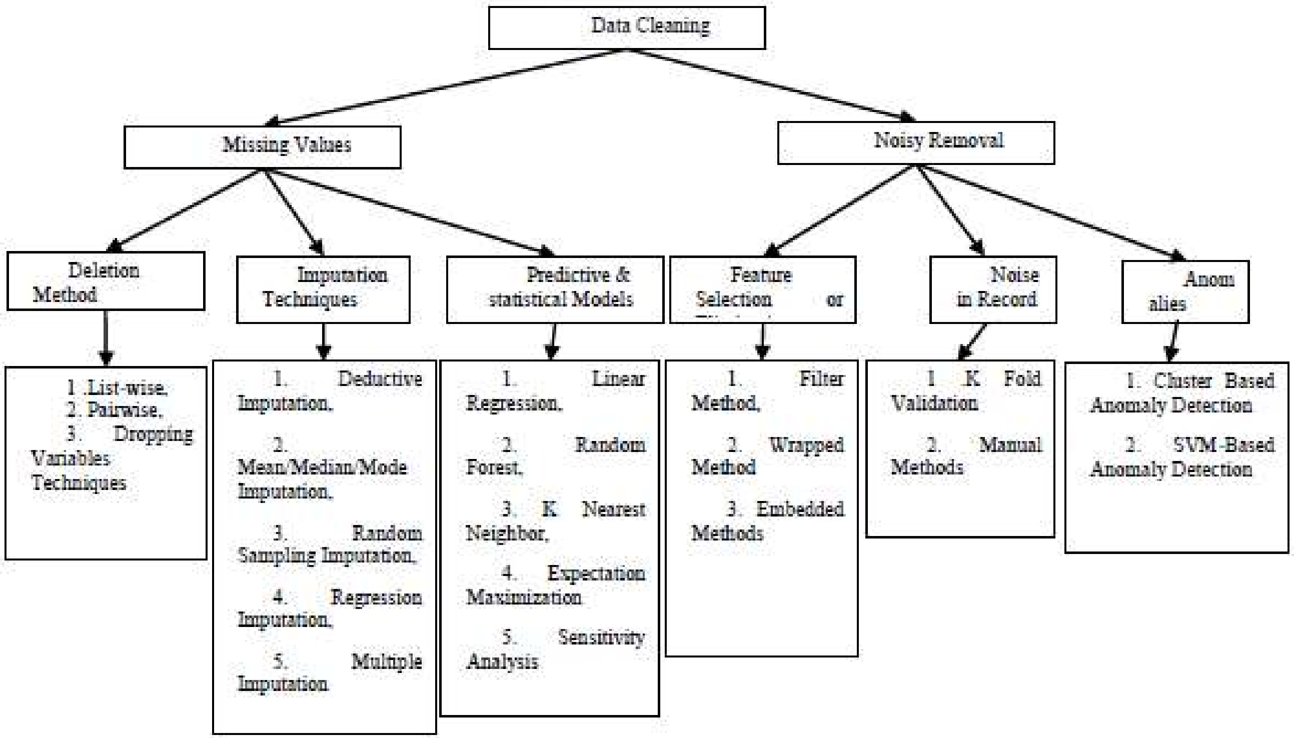
- **Problems with data**
  - dealing with noisy, inaccurate, uncertain and real-time data.
  - IoT data may contain missing and incomplete values that lead to poor data quality.
  - a lot of ambiguous information, and these data are large in size
  - constrained nature of IoT sensors and intermittent loss of connectivity, the massive scale of IoT data contains more irregularities and uncertainties,
- Preprocessing is **ensuring completeness in IoT data** that leads the data quality.
- It **removes the irregularities and uncertainties** in the data.
- before mining or modeling the data, it must be passed through improvement techniques known as data preprocessing.

Different data pre-processing techniques



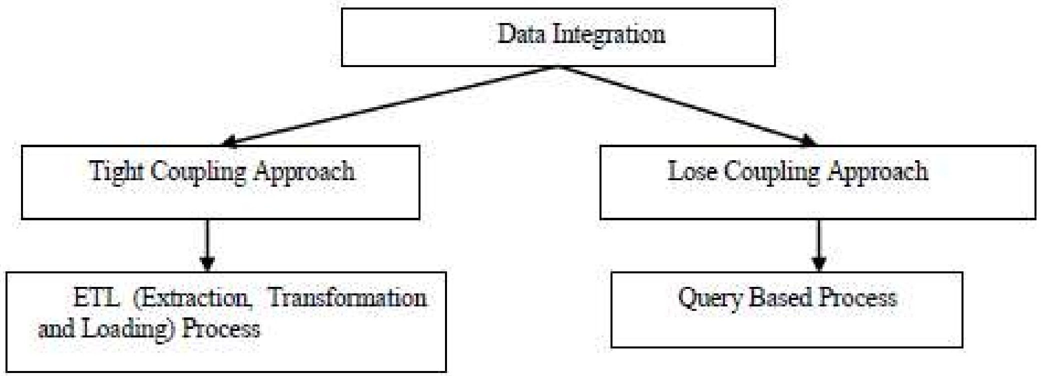
Data cleaning

- the process of eliminating the erroneous and missing part in the data.
- The process of handling these noisy and missing values can be achieved by various ways



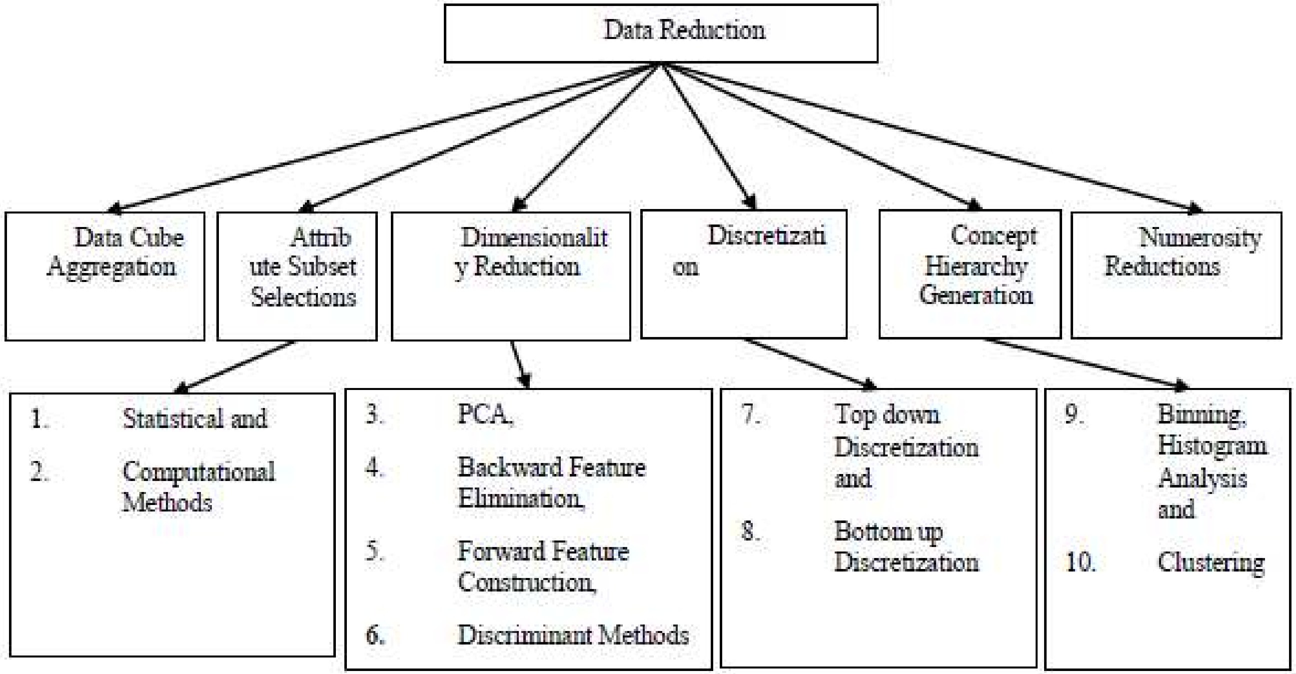
Data Integration

- It combines data from different source and giving users an integrated view of this data. Mainly, Data integration is done through two main approaches



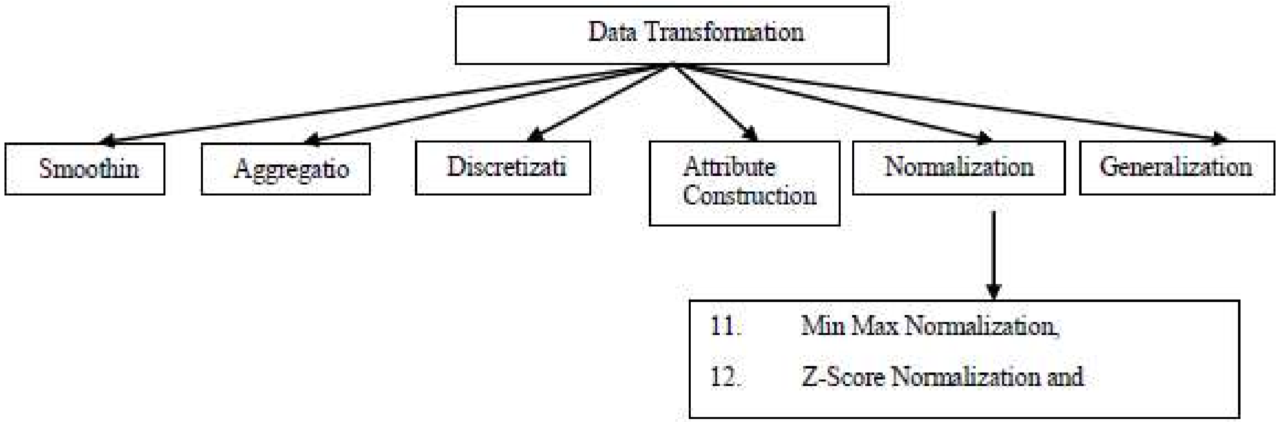
Data reduction

- Data reduction approaches utilized to diminish the unnecessary data as well as improve analytical process



Data Transformation

- the process of converts' data from one format to another format. Data transformation includes various functions to achieve the perfect format



Steps in Data Cleaning Process

What is Data cleaning

- Applying different techniques based on the problem and the data type.
- Incorrect data is either removed, corrected, or imputed.

Different Steps performed in Data Cleaning

- **Duplicates,**
- **Type conversion,**
- **Syntax errors,**
- **Standardize,**
- **Scaling / Transformation,**
- **Normalization,**
- **Missing Values,**
- **Outlier Treatment,**
- **Irrelevant Data.**

Duplicates (step 1 of 9)

- Duplicates are data points that are repeated in your dataset.
- It often happens when for example; Data are combined from different sources

Type Conversion (step 2 of 9)

- Make sure numbers are stored as numerical data types.
- A date should be stored as a date object, or a Unix timestamp (number of seconds), and so on.
- Categorical values can be converted into and from numbers if needed.

Syntax Errors (step 3 of 9)

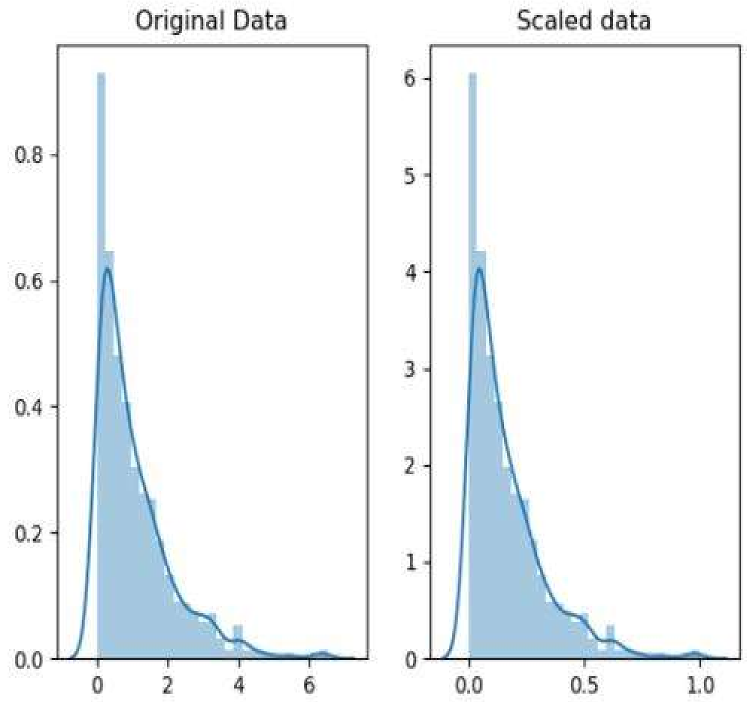
- **Remove white spaces:** Extra white spaces at the beginning or the end of a string should be removed.
- **Pad strings:** Strings can be padded with spaces or other characters to a certain width.
  - For example, some numerical codes are often represented with prepending zeros to ensure they always have the same number of digits.
- **Fix typos:** Strings can be entered in many different ways, and no wonder, can have mistakes.

Standardize (step 4 of 9)

- Our duty is to not only recognize the typos but also put each value in the same standardized format.
- **For strings,** make sure all values are either in lower or upper case.
- **For numerical values,** make sure all values have a certain measurement unit.

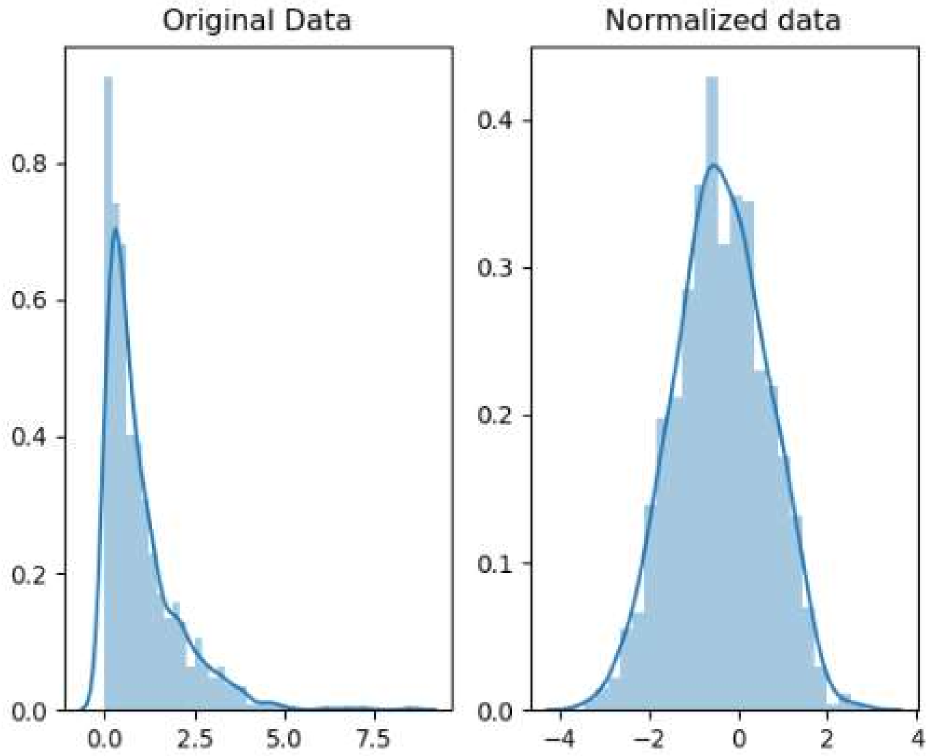
Scaling & Transformation (step 5 of 9)

- Scaling means to transform your data so that it fits within a specific scale, such as 0–100 or 0–1.
  - **For example**, exam scores of a student can be re-scaled to be percentages (0–100) instead of GPA (0–10).
- It can also help in making certain types of data easier to plot.



Normalization (step 6 of 9)

- In most cases, we normalize the data if we're going to be using statistical methods that rely on normally distributed data.



Missing Values (step 7 of 9)

- **Ways to Resolve Missing Values are:**
  - **Drop:** If the missing values in a column rarely happen and occur at random, then the easiest and most forward solution is to drop observations (rows) that have missing values.
  - **Impute:** It means to calculate the missing value based on other observations. There are quite a lot of methods to do that.
    - **Types : Statistical values**; **linear regression; Hot-Deck.**
  - **Flags:** Mark the missing values and flag them out for later analysis

Outlier Treatment (step 8 of 9

- They are values that are significantly different from all other observations.
  – Any data value **lies more than (1.5 * IQR) away from the Q1 and Q3** quartiles is considered an outlier.

Irrelevant Data (step 9 of 9)

- Irrelevant data are those that are not actually needed, and don’t fit under the context of the problem we’re trying to solve.
  - **For example**, If we were analyzing data about the general health of the population, the phone number wouldn’t be necessary — **column-wise**.

Other steps (Data cleaning)

Apply one of the following methods to solve this problem:

- **Binning**. Use binning if you have a pool of sorted data. **Divide all the data into smaller se**gments of the same size and apply your dataset preparation methods separately on each segment. For example, you can *bin* the values for Age into categories such as 21-35, 36-59, and 60-79.
- **Regression.** Regression analysis helps **to decide what variables do indeed have an impact**. Apply regression analysis to smooth large volumes of data. This will allow you to only work with the key features instead of trying to analyze an overwhelming amount of variables.
- **Clustering.** clustering algorithms to group the data.

Pre-processing 2: Data transformation

- By cleaning and smoothing the data, have already performed data modification.
- Data transformation are the methods of **turning the data into an appropriate format.**

- **Aggregation**
  - the data is pooled together and presented in a unified format for data analysis.
    - Working with a large amount of high-quality data allows for getting more reliable results from the ML model.
- **Normalization**
  - Normalization helps you to **scale the data within a range to avoid building incorrect ML models** while training and/or executing data analysis.
  - Why?
    - If the data range is very wide, it will be **hard to compare the figures**.
    - With various normalization techniques, you can transform the original data linearly, perform decimal scaling or Z-score normalization.
- **Feature selection**
  - It is the process of automatically choosing relevant features for your machine learning model want to predict
  - Feature selection is **the selection of variables** in data that are the **best predictors for the output variable**

- **Discretization**
  - During discretization, a programmer transforms the data into sets of small intervals. For example, putting people in categories “young”, “middle age”, “senior” rather than working with continuous age values. Discretization helps to improve efficiency.
- **Concept hierarchy generation**
  - If you use the concept hierarchy generation method, you can generate a hierarchy between the attributes where it was not specified.
    - For example, if you have the location information that includes a street, city, province, and country but they have no hierarchical order, this method can help you transform the data.

Data Modelling

- Dataset→preprocessing→ML modeling(categorization of data)→Analysis of data(understanding)→testing→deployment.

Data analyzation

- Analyzing the **mass amount of dat**a in the most efficient manner possible is the umbrella of data analytics.
- IoT data analytics refers to the analysis of every fragment of data generated from IoT devices at right time in order to extract intelligent insights
  - Not all data is the same; it can be categorized and analyzed in different ways.
- Depending on how data is categorized, various data analytics tools and processing methods can be applied.
  - Two important categorizations from an IoT perspective are whether the data is **structured or unstructured** and whether it is in **motion or at rest.**

Different categories of IoT data

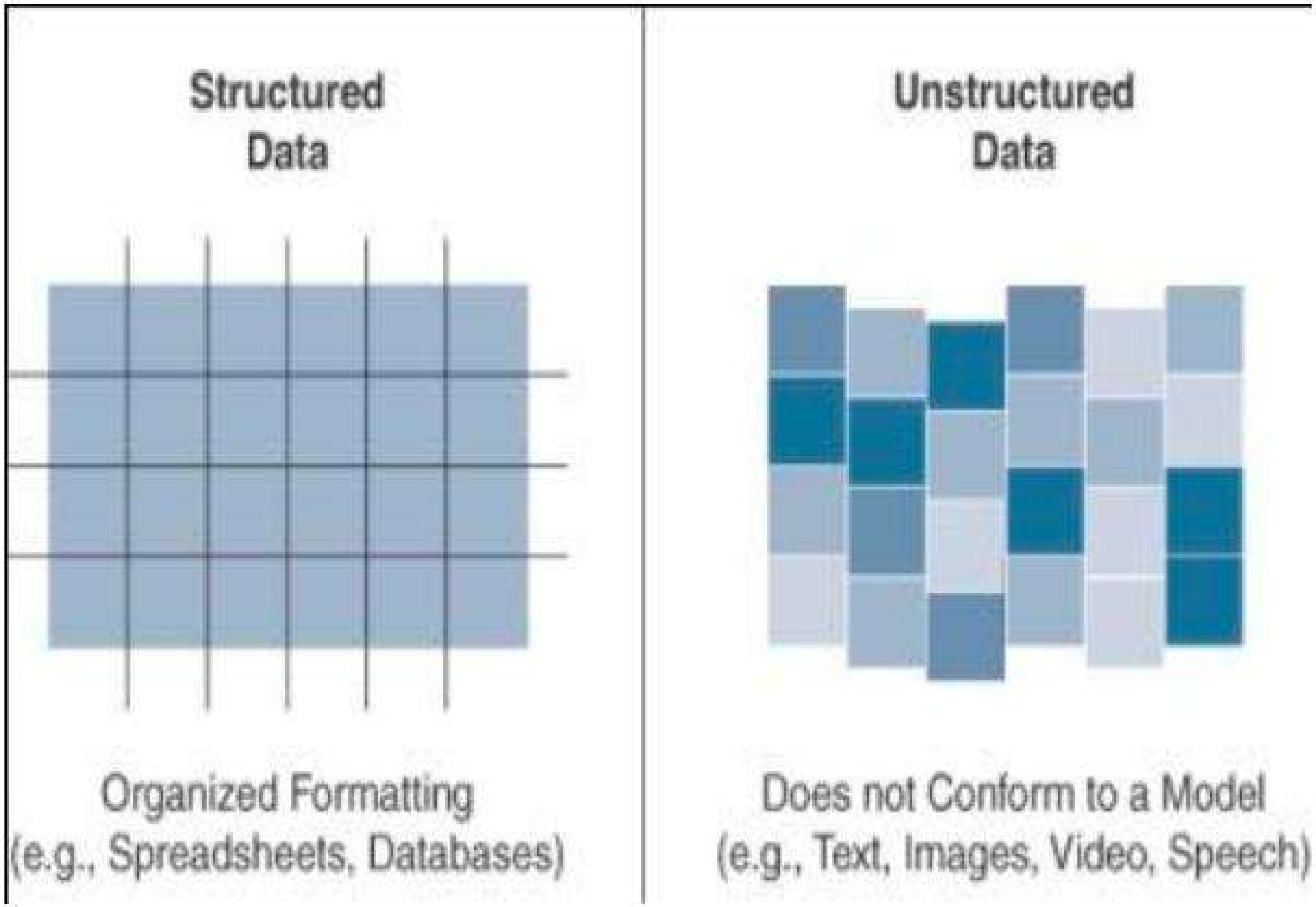
Structured data

- data follows a model or schema that defines how the data is represented or organized, meaning it fits well with a traditional relational database management system (RDBMS).
- structured data in a simple tabular form—for example, a spreadsheet where data occupies a specific cell and can be explicitly defined and referenced
- can be found in most computing systems and includes everything from banking transaction and invoices to computer log files and router configurations
- Structured data is easily formatted,stored,queried,and processed
- IoT sensor data often uses structured values, such as temperature, pressure, humidity, and so on, which are all sent in a known format.
- wide array of data analytics tools are readily available for processing this type of data scripts to commercial software like Microsoft Excel and Tableau

Unstructured data

- Unstructured data lacks a logical schema for understanding and decoding the data through traditional programming means.
- Examples of this data type include text, speech, images, and video.
- any data that does not fit neatly into a predefined data model.
- data analytics methods that can be applied to unstructured data, such as cognitive computing and machine learning
- With machine learning applications, such as natural language processing (NLP), you can decode speech.
- With image/facial recognition applications, you can extract critical information from still images and video

Different categories of IoT data



Different categories of IoT data

Data in motion

- Data in IoT networks is in transit
  - Data in motion is often streamed continuously from IoT devices to data processing and analytics platforms. This streaming data can include sensor readings, telemetry data, event logs, or any other information generated by IoT devices
- the data from smart objects is considered data in motion as it passes through the network and route to its final destination : edge or fog computing
- traditional client/server exchanges, such as web browsing and file transfers, and email.
- Tools supported: Spark, Storm, and Flink

Data at rest

- Data in IoT networks is being held or stored.
- Data saved to a hard drive, storage array, or USB drive
- Typically found in IoT brokers or in some sort of storage array at the data center

What kind of data it is?

- in a smart city application, data include real-time traffic flow data from vehicle sensors or environmental sensor readings from air quality monitoring stations.
- Is it data in rest or motion?

Application of analytics

- Descriptive analysis
- Diagnostic analysis
- Predictive analysis
- Prescriptive analysis

Descriptive analysis

- delineates **what has occurred** and **what is going on.**
- tells what is happening, either now or in the past.
  - For example, a thermometer in a truck engine reports temperature values every second.
- It is used to pull the data at any moment to gain insight into the current operating condition of the truck engine.
  - If the temperature value is too high, then there may be a **cooling problem** or the engine may be experiencing too much load.

Diagnostic Analysis

- Interested the problems for the questions of : **“Why”.**
- the example of the temperature sensor in the truck engine, you might wonder why the truck engine failed.
  - Diagnostic analysis might show that the temperature of “**the engine was too high, and the engine overheated”**.
- Applying diagnostic analysis across the data generated by a wide range of smart objects can provide a clear picture of why a problem or an event occurred

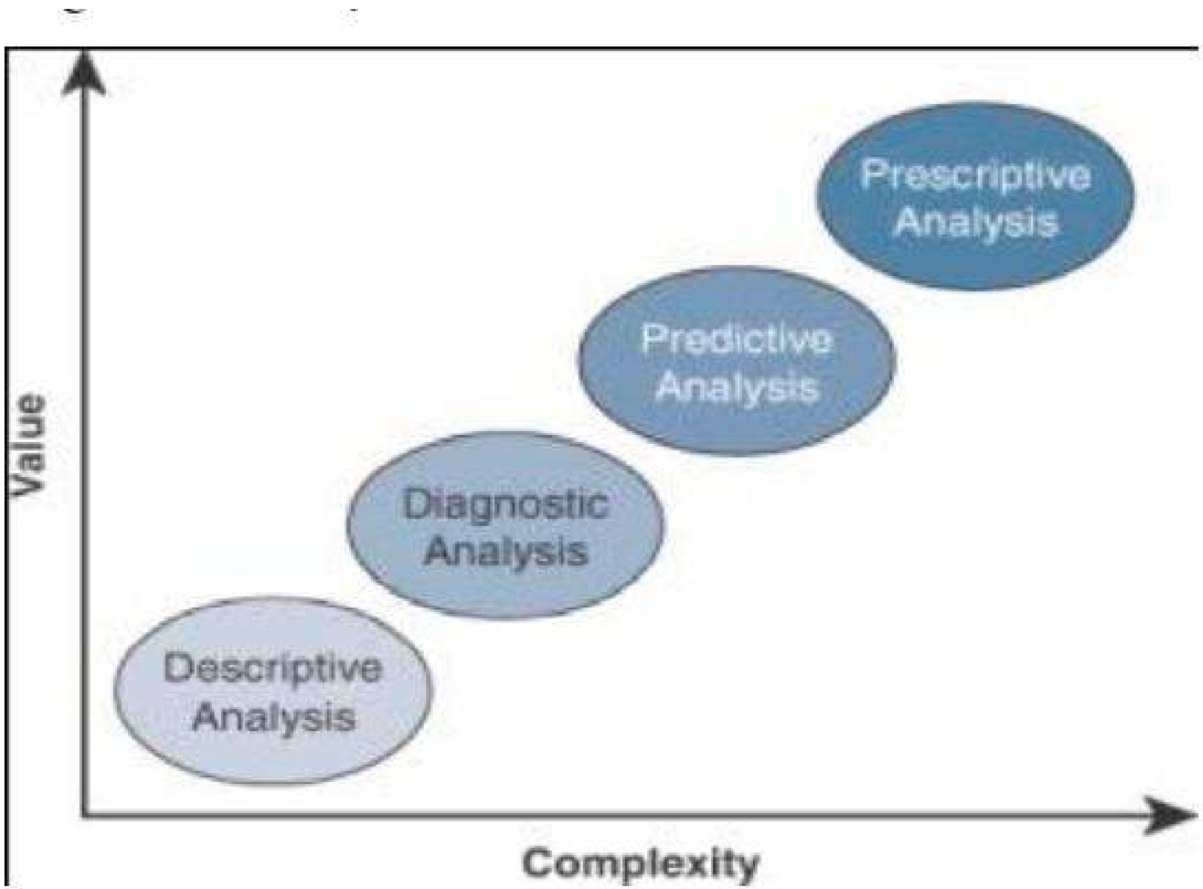
Predictive Analysis

- It aims to foretell problems or issues before they occur.
  - For example, with historical values of temperatures for the truck engine, predictive analysis could provide **an estimate on the remaining life of certain components in the engine**.
- These components could then be proactively replaced before failure occurs.
- Or perhaps if temperature values of the truck engine start to rise slowly over time, this could indicate the need for an oil change or some other sort of engine cooling maintenance

Prescriptive Analysis

- predictive and recommends solutions for upcoming problems.
- A prescriptive analysis of the temperature data from a truck engine might **calculate various alternatives to cost-effectively maintain our truck**
- These calculations could range from the cost necessary for more frequent oil changes and cooling maintenance to installing new cooling equipment on the engine or upgrading to a lease on a model with a more powerful engine.
- Prescriptive analysis looks at a variety of factors and makes the appropriate recommendation

Complexity vs Value of Analysis



Descriptive analytics on IoT data (what)

- Focuses on **what's happening,** by monitoring the status of IoT devices, machines, products and assets.
- Determines if things are going as planned, and notifies if anomalies occur.
- Descriptive analytics is generally **implemented as dashboards that show current and historical sensor data, key performance indicators (KPIs), statistics and alerts**.
- Addresses questions such as:
  - **Are there any anomalies that demand attention?**
  - **What's the utilization and throughput of this machine?**
  - **How are consumers using our products?**
  - **Where do my assets reside?**
  - **How many components are we creating with this tool?**
  - **How much energy is this machine using?**

Uses of Descriptive Analysis

- Descriptive methodologies focus on analyzing historic data for the purpose of **identifying patterns or trends.**
- Analytic techniques that fall into this category are most often associated with **exploratory data analysis which identifies central tendencies, variations, and distributional shapes**.
- Descriptive methodologies can also **search for underlying structures within data when there is no *priori* knowledge about patterns and relationships** are assumed.
- This can include correlation analysis, exploratory factor analysis, principal component analysis, trend analyses, and cluster analysis.

Descriptive Analysis in R Programming

- In Descriptive analysis, we **describing our data** with the help of various representative methods like using **charts, graphs, tables, excel files, etc.**
- Present it in a meaningful way so that it can be easily understood.
- Some measures that are used to describe a data set are
  - **Measures of central tendency and**
  - **Measures of variability or dispersion.**

Measure of central tendency

- It represents the whole set of data by a single value.
- It gives us the location of central points.
- There are three main measures of central tendency:
  - **Mean**
  - **Mode**
  - **Median**

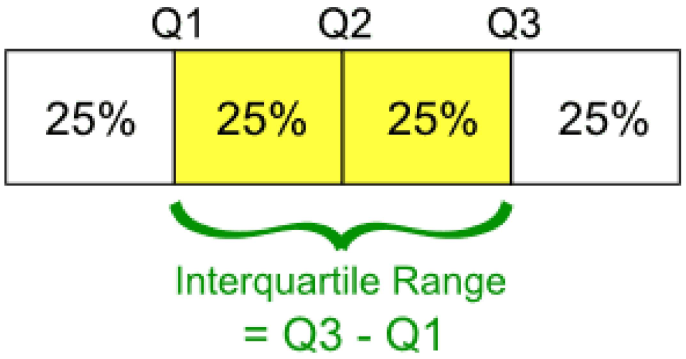
Measure of variability

- Measure of variability is known as the spread of data or how well data is distributed.
- The most common variability measures are:
  - **Range**
  - **Variance**
  - **Standard deviation**

Analysis	R Function
Mean	mean()
Median	median()
Mode	mfv() [modeest]
Range of values (minimum and maximum)	range()
Minimum	min()
Maximum	maximum()
Variance	var()
Standrad Deviation	sd()
Sample quantiles	quantile()
Interquartile range	IQR()
Generic function	summary()

Mean	It is the sum of observations divided by the total number of observations. (average which is the sum divided by count)
Median	It is the middle value of the data set. It splits the data into two halves. If the number of elements in the data set is odd then the center element is median and if it is even then the median would be the average of two central elements.
Mode	It is the value that has the highest frequency in the given data set. The data set may have no mode if the frequency of all data points is the same. Also, can have more than one mode if we encounter two or more data points having the same frequency.
Range	The range describes the difference between the largest and smallest data point in our data set. The bigger the range, the more is the spread of data and vice versa. **Range = Largest data value – smallest data value**

Variance	It is defined as an average squared deviation from the mean. It is being calculated by finding the difference between every data point and the average which is also known as the mean, squaring them, adding all of them, and then dividing by the number of data points present in data set.
Standard Deviation	It is defined as the square root of the variance. It is being calculated by finding the Mean, then subtract each number from the Mean which is also known as average and square the result. Adding all the values and then divide by the no of terms followed the square root.
Quartiles	A quartile is a type of quantile. The first quartile (Q1), is defined as the middle number between the smallest number and the median of the data set, the second quartile (Q2) – the median of the given data set while the third quartile (Q3), is the middle number between the median and the largest value of the data set.



Interquartile Range	The interquartile range (IQR), also called as midspread or middle 50%, or technically H-spread is the difference between the third quartile (Q3) and the first quartile (Q1). It covers the center of the distribution and contains 50% of the observations. **IQR = Q3 – Q1**
summary()	The function summary() can be used to display several statistic summaries of either one variable or an entire data frame.

Different forms of Descriptive Analytics

Common forms of Descriptive analytics

Numerical data descriptive statistics

	mpg	hp	wt
median	19.200	123.000	3.325
mean	20.091	146.688	3.217
SE.mean	1.065	12.120	0.173
CI.mean.0.95	2.173	24.720	0.353
var	36.324	4700.867	0.957
std.dev	6.027	68.563	0.978
coef.var	0.300	0.467	0.304

Categorical data descriptive statistics

		BC	CA	DF
Married	Female	190	638	188
	Male	197	692	210
Single	Female	183	686	175
	Male	239	717	242

Classical Analyses

- Numerical data descriptive statistics
- Categorical data descriptive statistics
- Assumption of normality
- Assumption of homogeneity
- Assessing correlations
- Univariate statistical inference
- Multivariate statistical inference
- Bootstrapping for parameter estimates

Text Mining

- Tidying Text & Word Frequency
- Sentiment Analysis
- Term vs. Document Frequency
- Word Relationships
- Converting Between Tidy and Non-tidy Formats

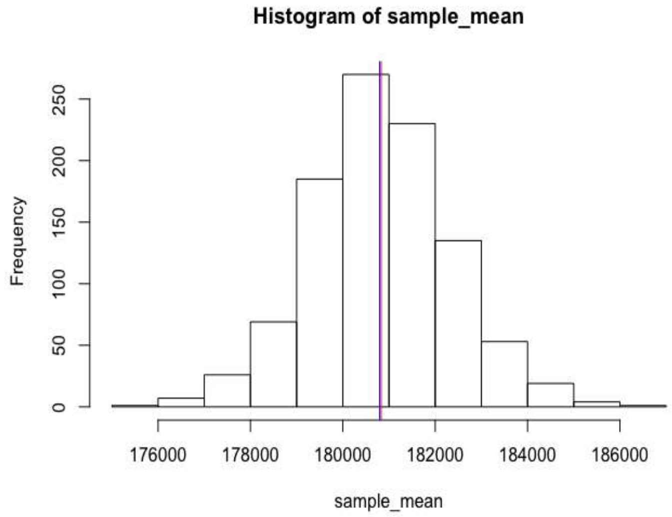
Unsupervised Learning

- Principal Component Analysis
- K-means Cluster Analysis
- Hierarchical Cluster Analysis

Univariate Statistical Inference- classical Analysis

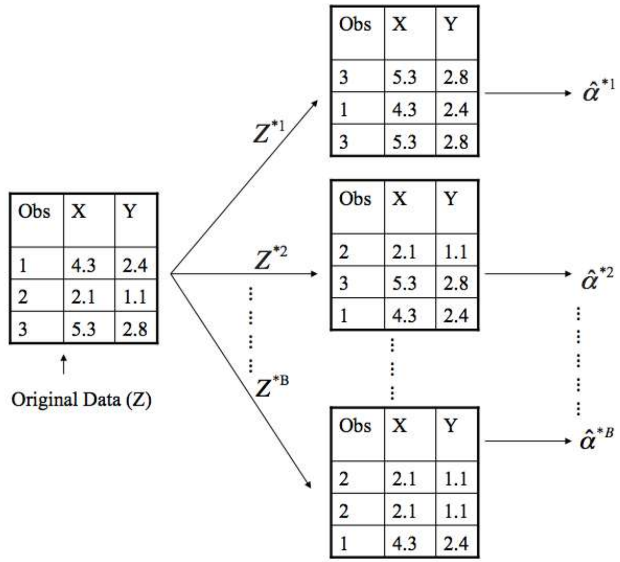
- univariate statistical inference, specifically in the context of classical analysis, refers to the process of making inferences or drawing conclusions about a single variable or population parameter based on sample data. This approach is fundamental in statistical analysis and hypothesis testing.

1. **Population and Sample**: In classical analysis, there's a clear distinction between the population and the sample. The population refers to the entire group or set of individuals, items, or observations of interest. The sample is a subset of the population selected for analysis. Univariate statistical **inference typically involves drawing conclusions about population parameters based on sample statistics**.
    1. Trying to know about sample population group whose HR values ranging between 70 to 80
    2. Population category whose HR values are >140 etc..
2. **Parameter Estimation**: One of the primary objectives in univariate statistical inference is to estimate population parameters based on sample data. For example, if we're interested in **estimating the population mean (μ) or variance (2σ2) of a variable, we use sample statistics such as the sample mean (¯x¯) or sample variance (2s2) as estimators.**
    1. Estimating the inferences for 100 IoT temperature sensors in a farming
        1. Taking 10 samples sensors data
        2. Calculates the mean or average of the temperature
        3. Using confidence intervals or hypothesis testing, the farm manager infers that the **average soil temperature** across the entire farm is **likely between 21.8°C and 23.2°C**, with 95% confidence.



Bootstrapping for Parameter Estimates-classical Analysis

- Resampling methods are an indispensable tool in modern statistics.
- They involve repeatedly drawing samples from a training set and re-computing an item of interest on each sample.
- *Bootstrapping* is one such resampling method that repeatedly draws independent samples from our data set and provides a direct computational way of assessing uncertainty.



Text Mining

- **Creating Tidy Text**
  - A fundamental requirement to perform text mining is to get text in a tidy format and perform word frequency analysis.
    - Text is often in an unstructured format so performing even the most basic analysis requires some re-structuring.
- **Sentiment Analysis**
  - To understand the opinion or emotion in the text from tidy text.

- **Term vs. Document Frequency**
  - Focused on identifying the frequency of individual terms within a document along with the sentiments that these words provide.
  - It is also important to understand **the importance that words provide within and across documents.**
    - *Term frequency* (tf) identifies how frequently a word occurs in a document.
    - That many common words such as “the”, “is”, “for”, etc. typically top the term frequency lists.
- One approach to correct for these common, yet low context words, is to remove these words by using a list of stop words.
- Another approach is to use what is called a term’s *inverse document frequency* (idf),
  - which decreases the weight for commonly used words and increases the weight for words that are not used very much in a collection of documents.

Principal Components Analysis

- Principal Component Analysis (PCA) involves the process by which principal components are computed, and their role in understanding the data.
- PCA is an unsupervised approach, which means that it is performed on a set of variables $X_1, X_2, \ldots, X_p$ with no associated response Y.
- PCA reduces the dimensionality of the data set, allowing most of the variability to be explained using fewer variables.
- PCA is commonly used as one step in a series of analyses.
- The goal of PCA is to explain most of the variability in the data with a smaller number of variables than the original data set.
- For a large data set with p variables, we could examine pairwise plots of each variable against every other variable, but even for moderate p, the number of these plots becomes excessive and not useful.

K-means Cluster Analysis

- Clustering is a broad set of techniques for finding subgroups of observations within a data set.
- When we cluster observations, we want observations in the same group to be similar and observations in different groups to be dissimilar.
- Because there isn’t a response variable, this is an unsupervised method, which implies that it seeks to find relationships between the n observations without being trained by a response variable.
- Clustering allows us to identify which observations are alike, and potentially categorize them therein.
- K-means clustering is the simplest and the most commonly used clustering method for splitting a dataset into a set of k groups.

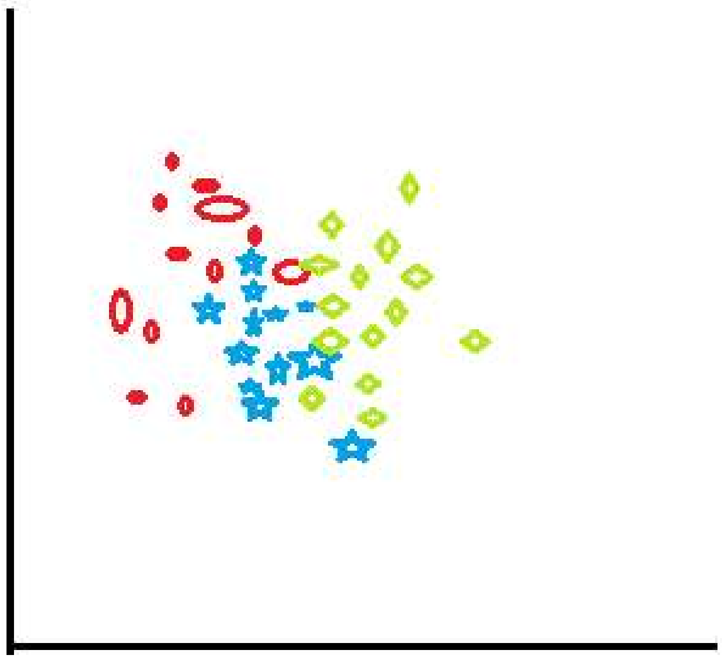


fig 1: before applying k-means clustering

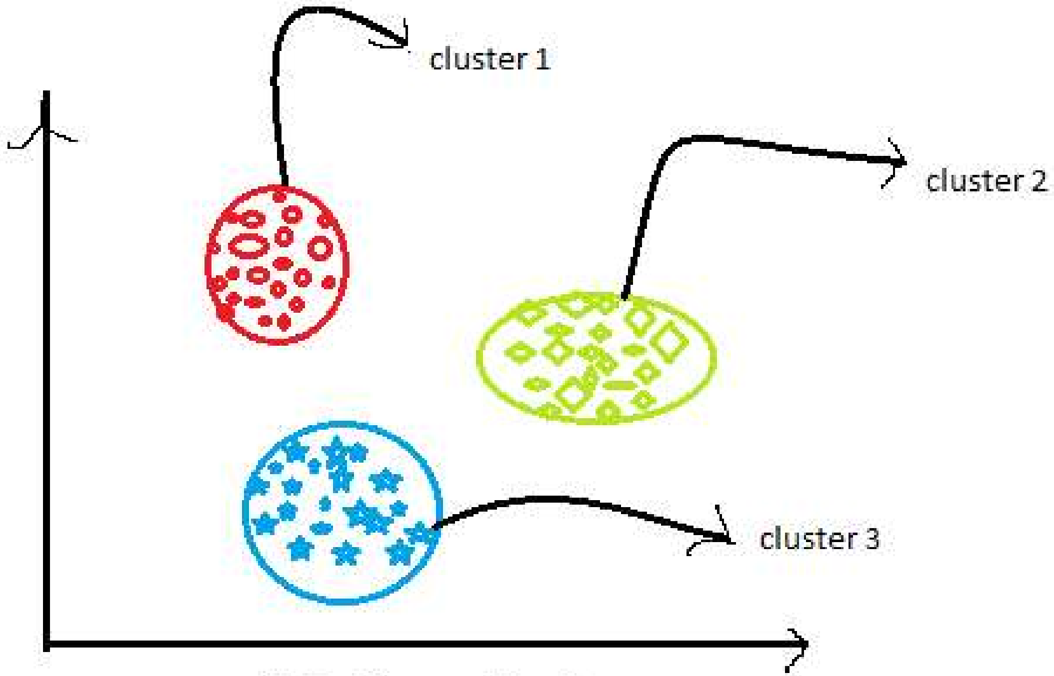
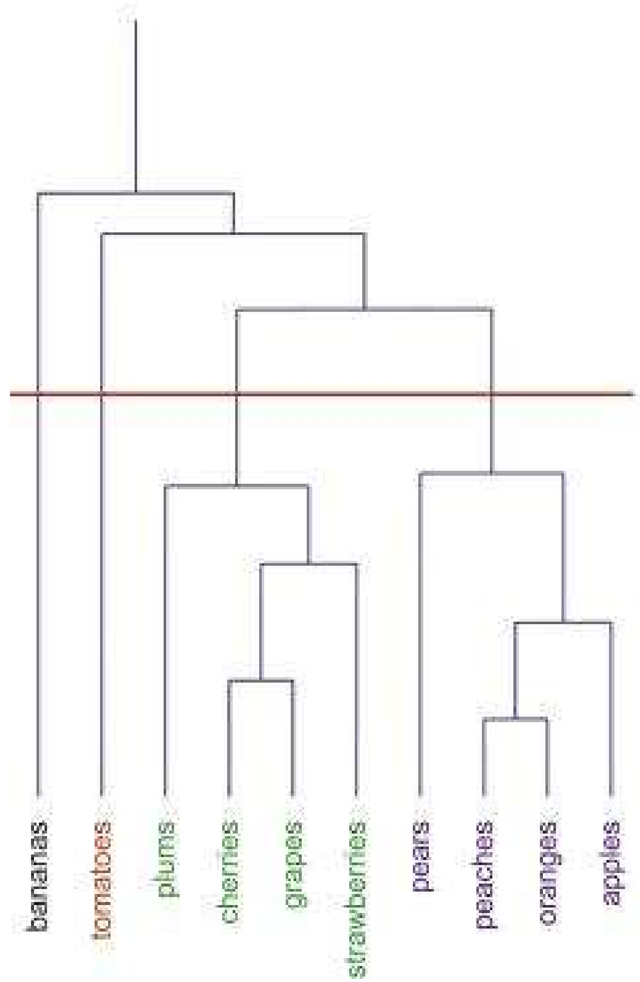


fig 2: After applying K-means clustering

Hierarchical Cluster Analysis

- Hierarchical clustering is an alternative approach to k-means clustering for identifying groups in the dataset.
- It does not require us to pre-specify the number of clusters to be generated as is required by the k-means approach.
- Furthermore, hierarchical clustering has an added advantage over K-means clustering in that it results in **an attractive tree-based representation of the observations**.



Scenario

- An XYZ mobile company who have their mobile v1 in the current market. Due to their heavy sales for the product they want to add some features and wants to release mobile v2. so the company wants to analyse willingness of their customer feedback. What kind of analytics is suitable here and justify it.

References

- https://www.geeksforgeeks.org/descriptive-analysis-in-r-programming/
- https://uc-r.github.io/descriptive

Diagnostic analytics on IoT data

- Answers the question: **why is something happening?**
- Analyzes IoT data **to identify core problems and to fix or improve a service, product or process**.
- Diagnostic capabilities are typically **extensions to dashboards** that permit users to **drill into data, compare it, and visualize correlations** and trends in an ad-hoc manner.
- Many organizations employ domain experts knowledgeable about a specific process, machine, device or product, rather than data scientists, to perform diagnostics on data.
- Addresses questions such as:
  - **Why is this machine producing more defective parts than other machines?**
  - **Why is this machine consuming excessive energy?**
  - **Why aren't we producing enough parts with this tool?**
  - **Why are we getting a lot of product returns from American customers?**

How Does Diagnostic Analytics Work?

- Diagnostic analytics uses a variety of techniques **to provide insights** into the causes of trends. These include:
- **Data drilling:**
  - Drilling down into a dataset can reveal more detailed information about which aspects of the data are driving the observed trends.
  - For example, analysts may drill down into **national sales data to determine whether specific regions, customers or retail** channels are responsible for increased sales growth.

- **Data mining**
  - hunts **through large volumes of data to find patterns and associatio**ns within the data.
    - For example, data mining might reveal the most common factors associated with a rise in insurance claims.
  - Data mining can be conducted manually or automatically with machine learning technology.
- **Correlation analysis**
  - examines how strongly different variables are linked to each other.
  - For example, sales of ice cream and refrigerated soda may soar on hot days.

Three Diagnostic Analytics Categories

- The diagnostic analytics process of determining the root cause of a problem or trend typically comprises three primary stages.
- **Identify anomalies:**
  - Trends or anomalies highlighted by descriptive analysis may require diagnostic analytics **if the cause isn't immediately obvious.**
    - you might notice something **unusual**—like a sudden drop in sales, a spike in website traffic, or a dip in performance.
  - In addition, it can sometimes be difficult to determine whether the results of descriptive analysis really show a new trend, especially if there's a lot of natural variability in the data.
  - In those cases, statistical analysis can help to determine whether the results actually represent a departure from the norm.
    - **Descriptive analysis:** "Sales dropped by 30% in March."
    - **Anomaly spotted:** That's unusual, and we don't know why.
    - **Diagnostic analytics:** You look into possible causes—maybe a competitor launched a new product, there were supply chain issues, or your ads weren't running

- **Discovery:**
  - **Data discovery** is like detective work—you're gathering clues from both inside and outside your company to **figure out why the unexpected thing happened**.
  - That may involve gathering external data (outside the organization) as well as drilling into internal data.
    - Identifying the associations of data

Examples of external data that might explain anomalies:

- **Supply chain issues** → maybe a vendor delayed shipments.
- **Regulatory changes** → a new law made something harder to sell.
- **Competition** → a new competitor just entered the market.
- **Weather** → a storm caused delays or changed consumer behavior.

- **Causal relationships:**
  - Further **investigation can provide insights into whether the associations in the data point to the true cause of the anomaly.**
    - **Understanding the identified features are correct cause of anomaly.**
  - The fact that two events correlate doesn’t necessarily mean one causes the other.
    - For example increasing of ice cream sales and umbrella sales for an e-commerce applications does not mean both are related for increase on sales, the sales increase is because of shot summer.
  - Deeper examination of the data associated with the sales increase can indicate which factor or factors were the most likely cause.

Benefits of Diagnostic Analytics

- Understanding the **causes of business outcomes** is critical to a company's ability to grow and learn from mistakes.
- Diagnostic analytics lets companies closely look into the factors that drive success or cause failure, including contributing factors that may not be obvious at first glance.
  - Let's say a company's sales suddenly jump: diagnostic analytics might reveal **other contributing factors**, like:
    - A competitor went out of stock
    - A positive news article about the brand
    - Changes in customer behavior
- Diagnostic analytics can help to **instill a data-driven analytical culture** throughout the business.
  - Making use **data to make smarter decisions**, solve problems, and move the business forward—every day.
- When business leaders understand that the company has the tools **to investigate the cause of problems**, they're more likely to use diagnostic analytics in their decision-making.

Drawbacks of Diagnostic Analytics

- A drawback of diagnostic analytics is that **it focuses on historical data;**
- it can only help **businesses understand why events happened in the past.**
- Further investigation may be needed to determine whether the correlations revealed by diagnostic analytics really show cause and effect.
- To look into the future, businesses need to use other analytic techniques, such as predictive analytics, which examines the potential future impact of trends and events, and
- Prescriptive analytics, which suggests actions businesses can take to influence the outcome of those future trends.

Example Scenario II

- Recently I installed a new software XYZ in my Laptop. After continuous notifications about LAPTOP restart, I restarted my LAPTOP. From next immediate restart I found some of file formats are not recognized and my processor become very slow. How will you analyse this problem using Descriptive and Diagnostic analytics.

References

- https://www.netsuite.com/portal/resource/articles/data-warehouse/diagnostic-analytics.shtml

Tools & technology supported

- Data Mining and Machine learning
- Big IoT data analytics,
  - creating smarter IoT by extracting unseen patterns,
  - hidden correlations, trends, inferences, and actionable insights, facilitating greater intelligence,
  - smarter decision making, enhancing performance, automation, productivity, and accuracy.

What is IoT Search Engines (IOTSE)

- It Is a search tool to search the list of IoT devices connected to the internet or search and identity recognition of IOT devices.
- This tool provides a set of information related to IOT devices like type, ports, operating system and location.
- This tool provides a huge amount of data.
- The data provided must be analyzed and interpreted in the aim to give a better understanding, quickly description and help to make decisions.
- It will solve the search issue of the internet of things

IoT Search Engines: importance of Exploratory Data Analysis

- Exploratory data analysis(EDA) on IOT search engine data in order to give an analysis of IOTSE data with visual method.
- Searching and identity visualization represent the essential factors which led to the development of IOTSE

IoTSE steps & tools

- data collection indexing and display of results
- most popular of this type are Shodan, Censys and Thingful

Issues of data interpretation of IOT search engine

- several challenges
  - interpretation of results, irrelevant results and analysis of queries and results
- Solution : exploratory data analysis can be used to analyze and make sense of data from the IOT search engine

Exploratory Data Analysis

Importance of exploratory data analysis?

- Exploratory data analysis (EDA) is used by data **scientists to analyze and investigate data sets and summarize their main characteristics, employing data visualization methods.**
- It helps determine how **best to manipulate data sources to get the answers need,** making it easier for data scientists to discover patterns, spot anomalies, test a hypothesis, or check assumptions.
- EDA is primarily used to see what data can reveal beyond the formal modeling or hypothesis testing task and provides a better understanding of data set variables and the relationships between them.
- It can also help determine if the statistical techniques you are considering for data analysis are appropriate.

Contd.,

- Data Scientists widely use EDA to understand datasets for decision-making and data cleaning processes.
- EDA reveals crucial information about the data, such as hidden patterns, outliers, variance, covariance, correlations between features.
- The information is essential for the hypothesis’s design and creating better-performing models.

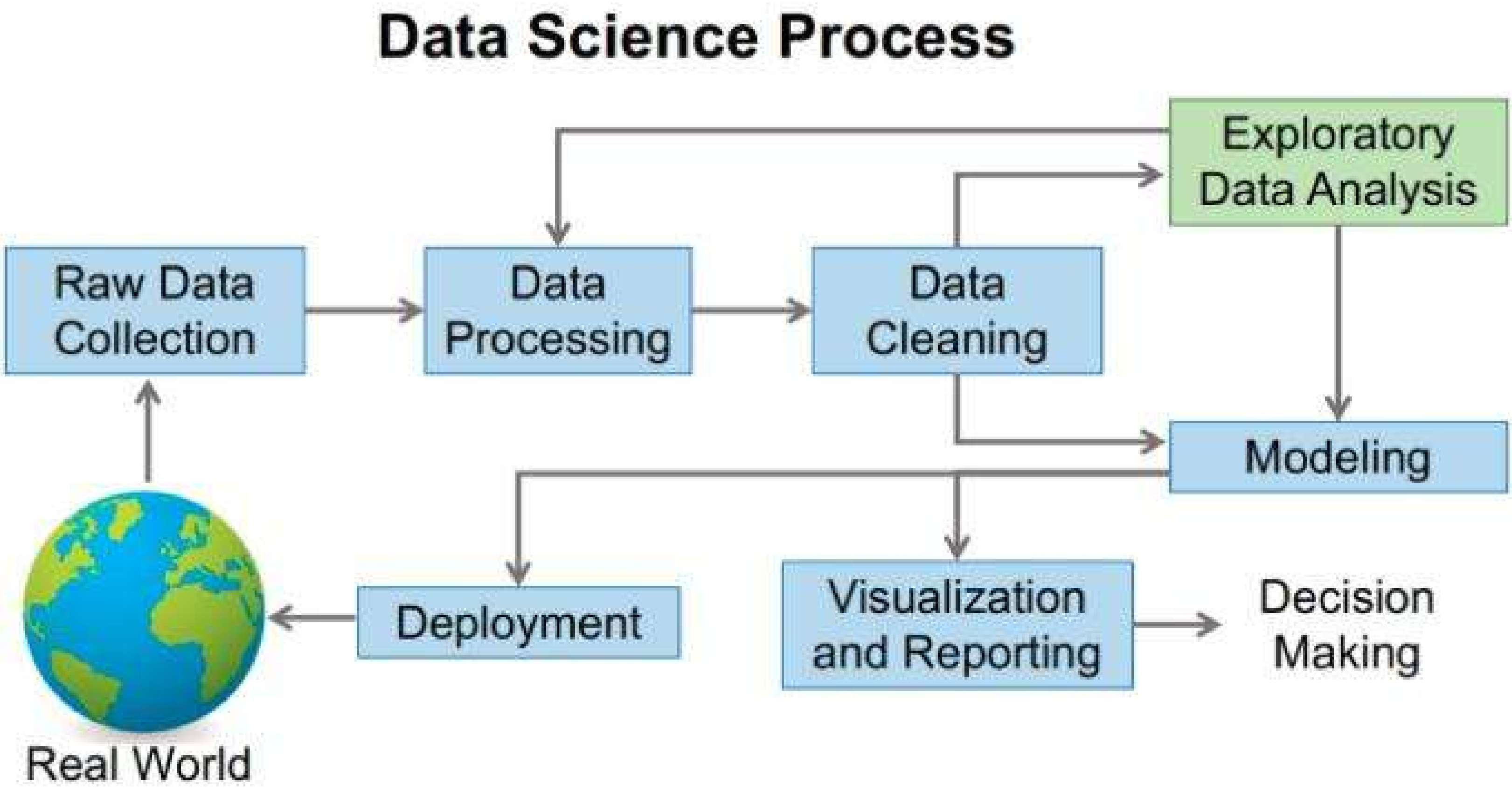
Definitions: Exploratory data analysis (EDA)

- It is an approach to analyzing data which serve to give a quick and understandable description of dataset.
- EDA is one of the step of data analysis process which consist of visualize data, taking the main characteristics and giving better understand by using visual methods. It’s an important approach to analyze expansive number of data like data generated in real time by IOT devices.
- It is an approach to analyzing datasets to summarize their main characteristics, often with visual methods

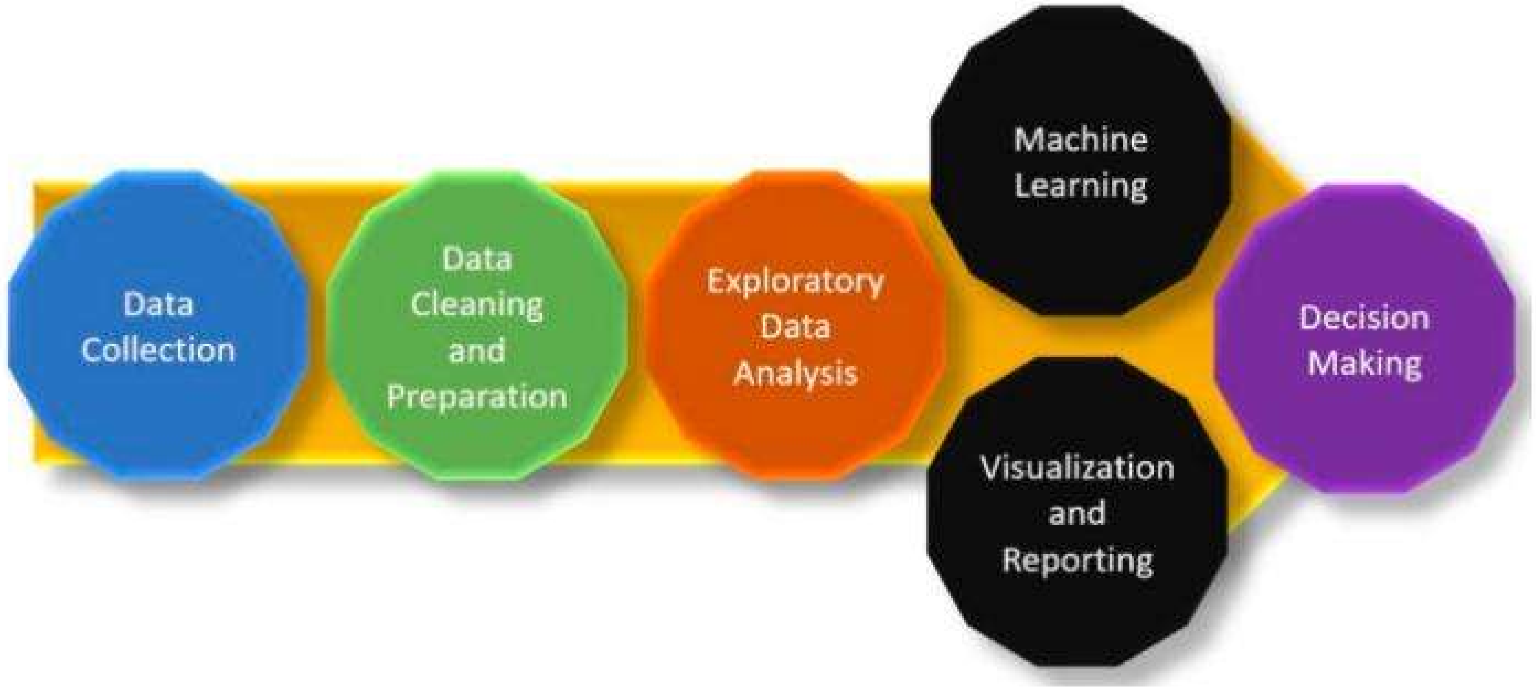
Why is exploratory data analysis important in data science?

- The main purpose of EDA is to help look at data **before making any assumptions.**
- It can **help identify obvious errors**, as well as better understand patterns within the data, detect outliers or anomalous events, find interesting relations among the variables.
- Data scientists can use exploratory analysis to ensure the results they produce are valid and applicable to any desired business outcomes and goals.
- EDA also **helps stakeholders by confirming they are asking the right questions.**
- EDA can help answer questions about **standard deviations,** categorical variables, and confidence intervals.
- Once EDA is complete and insights are drawn, its features can then be used for more sophisticated data analysis or modeling, including machine learning.

Process of EDA



The position of EDA in ML modelling



Analysis by EDA

- Data exploration using numerical analysis
- Data exploration using visual analysis
  - Preview data
  - Check total number of entries and column types
  - Check any null values
  - Check duplicate entries
  - Plot distribution of numeric data (univariate and pairwise joint distribution)
  - Plot count distribution of categorical data
  - Analyse time series of numeric data by daily, monthly and yearly frequencies

Exploratory data analysis tools

- Specific statistical functions and techniques to perform with EDA tools include:
  - **Clustering and dimension reduction techniques**, which help create graphical displays of high-dimensional data containing many variables.
  - **Univariate visualization** of each field in the raw dataset, with summary statistics.
  - **Bivariate visualizations and summary statistics** that allow you to assess the relationship between each variable in the dataset and the target variable you're looking at.
  - **Multivariate visualizations**, for mapping and understanding interactions between different fields in the data.

- **K-means Clustering** is a clustering method in unsupervised learning where data points are assigned into K groups, i.e. the number of clusters, based on the distance from each group's centroid.
- The data points closest to a particular centroid will be clustered under the same category.
- K-means Clustering is commonly used in market segmentation, pattern recognition, and image compression.
- **Predictive models**, such as linear regression, use statistics and data to predict outcomes.

EDA categories

- **It falls into two categories**
  - *The **univariate analysis** involves analyzing one feature, such as summarizing and finding the feature patterns.*
  - *The **multivariate analysis** technique shows the relationship between two or more features using cross-tabulation or statistics.*

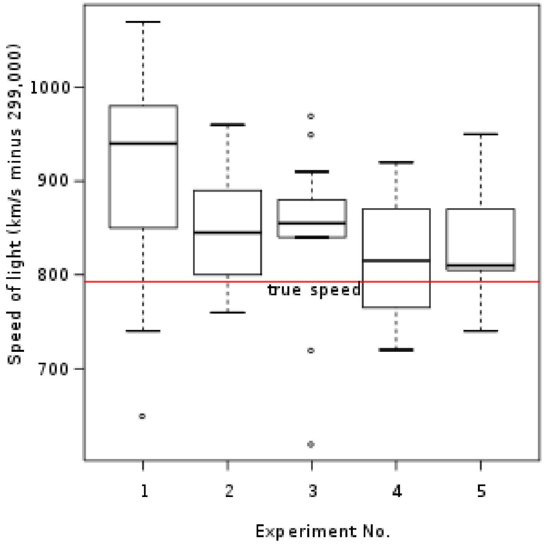
Types of exploratory data analysis

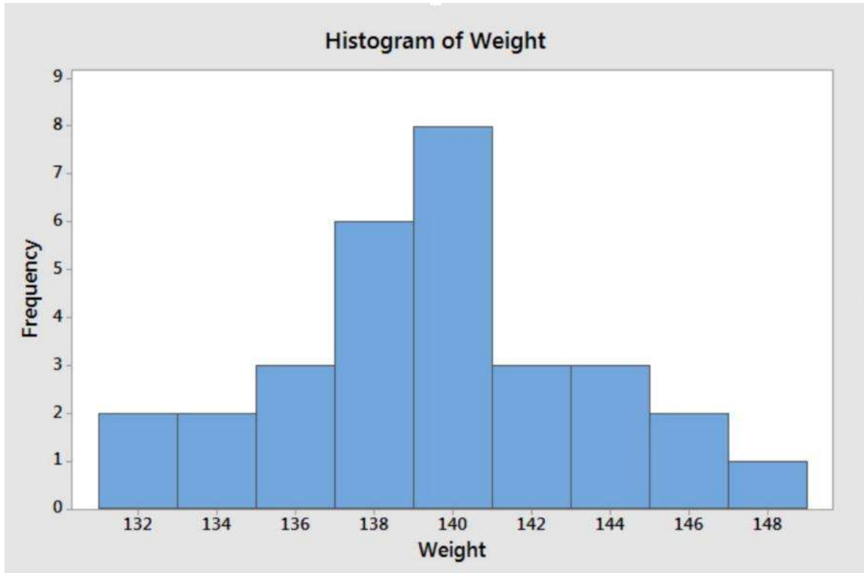
There are four primary types of EDA:

- **Univariate non-graphical.**
- This is simplest form of data analysis, where the data being analyzed consists of just one variable.
- Since it’s a single variable, it doesn’t deal with causes or relationships.
- The main purpose of univariate analysis is to describe the data and find patterns that exist within it.
- **Univariate graphical.** Non-graphical methods don’t provide a full picture of the data. Graphical methods are therefore required. Common types of univariate graphics include:
  - **Stem-and-leaf plots,** which show all data values and the shape of the distribution.
  - **Histograms**, a bar plot in which each bar represents the frequency (count) or proportion (count/total count) of cases for a range of values.
  - **Box plots**, which graphically depict the five-number summary of minimum, first quartile, median, third quartile, and maximum.

```
 2 | 0
 3 | 025
 4 | 11378
 5 | 133467889
 6 | 024559
 7 | 147
 8 | 8
 9 |
10 | 2
```

A stem-and-leaf plot of the values 20, 30, 32, 35, 41, 41, 43, 47, 48, 51, 53, 53, 54, 56, 57, 58, 58, 59, 60, 62, 64, 65, 65, 69, 71, 74, 77, 88 and 102





- **Multivariate nongraphical:** Multivariate data arises from more than one variable.
- Multivariate non-graphical EDA techniques generally show the relationship between two or more variables of the data through cross-tabulation or statistics.
- **Multivariate graphical:** Multivariate data uses graphics to display relationships between two or more sets of data.
- The most used graphic is a grouped bar plot or bar chart with each group representing one level of one of the variables and each bar within a group representing the levels of the other variable.

Other common types of multivariate graphics include:

- **Scatter plot,** which is used to plot data points on a horizontal and a vertical axis to show how much one variable is affected by another.
- **Multivariate chart,** which is a graphical representation of the relationships between factors and a response.
- **Run chart,** which is a line graph of data plotted over time.
- **Bubble chart,** which is a data visualization that displays multiple circles (bubbles) in a two-dimensional plot.
- **Heat map,** which is a graphical representation of data where values are depicted by color.

Reference

- https://www.ibm.com/cloud/learn/exploratory-data-analysis

References

- https://mite.ac.in/wp-content/uploads/2021/04/iot_module4.pdf.
- Tausifa Jan Saleem, Mohammad Ahsan Chishti, “Data Analytics in the Internet of Things: A Survey”, Article *in* Scalable Computing · December 2019.
- Erwin Adi, Adnan Anwar, Zubair Baig, “Sherali Zeadally3Machine learning and data analytics for the IoT”, 11 Mayhttps://towardsdatascience.com/exploratory-data-analysis-eda-a-practical-guide-and-template-for-structured-data-abfbf3ee3bd9 2020.
- https://www.digiteum.com/iot-data-collection/